MATH 241: Calculus III

Fall 2026

Lecture 3: Lines and Planes in Space

Background & Motivation

With vectors in hand, we can now describe geometric objects in 3D. A line needs a point and a direction; a plane needs a point and a normal. These are the simplest objects in space, and the techniques here—parametric equations, dot products for angles, cross products for normals—come back throughout the course.

1. Lines in $\mathbb{R}^3$

Definition 1: Parametric Equation of a Line

A line through $P_0 = (x_0, y_0, z_0)$ with direction vector $\mathbf{d} = \langle a, b, c \rangle$:

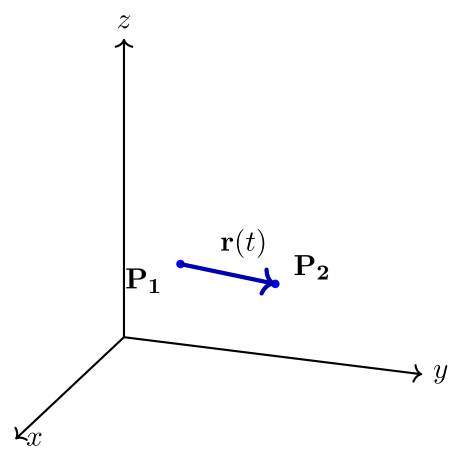
Vector form:

$$\mathbf{r}(t) = \_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_$$

Component form:

$$x = \_\_\_\_\_\_\_\_\_\_, \quad y = \_\_\_\_\_\_\_\_\_\_, \quad z = \_\_\_\_\_\_\_\_\_\_$$

– Each value of $t \in (-\infty, \infty)$ gives a point on the line



The line segment connecting points $\mathbf{P_1}$ and $\mathbf{P_2}$ in 3D, represented by $\mathbf{r}(t) = (1-t)\mathbf{P_1} + t\mathbf{P_2}$ for $t \in [0, 1]$.

Definition 2: Line Segment

The segment from P_1 to P_2:

$$\mathbf{r}(t) = \underline{\hspace{6cm}}$$

– $t = 0$ gives P_1; $t = 1$ gives P_2

Example 1: Parametric Line

Write the parametric equation of the line through $P_0 = (1, -2, 3)$ in direction $\mathbf{d} = \langle 4, -1, 2 \rangle$.

2. Planes in $\mathbb{R}^3$

Definition 3: Equation of a Plane

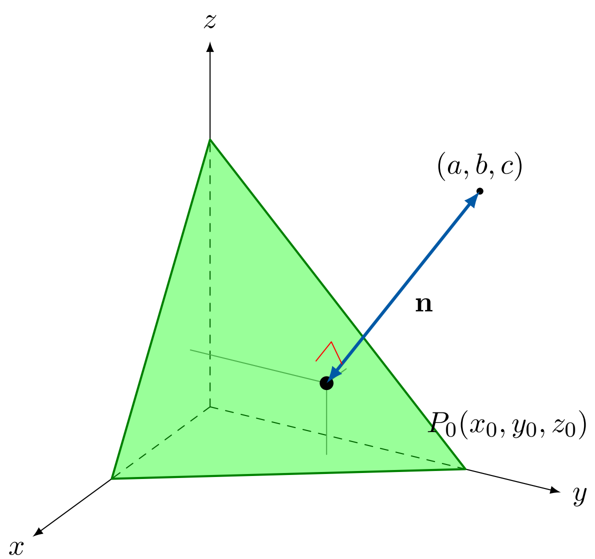
A plane through $P_0 = (x_0, y_0, z_0)$ with normal vector $\mathbf{n} = \langle A, B, C \rangle$:
Point-normal form:

$$\underline{\hspace{6cm}}$$

General form:

$$\underline{\hspace{6cm}}$$

where $D = Ax_0 + By_0 + Cz_0$



A plane in 3D space defined by a point P_0 and a normal vector $\mathbf{n}$ extending from P_0 in the perpendicular direction.

Example 2: Plane from Point and Normal

Find the equation of the plane through $(2, -1, 3)$ with normal $\mathbf{n} = \langle 4, 5, -2 \rangle$.

Example 3: Plane from Three Points

Find the equation of the plane through $P_1 = (1, 2, 3)$, $P_2 = (4, -1, 2)$, $P_3 = (-2, 3, 5)$.

3. Intersections and Angles

Line–Plane Intersection

– Strategy: substitute parametric equations of the line into the plane equation, solve for t

Example 4: Line–Plane Intersection

Find where $\mathbf{r}(t) = \langle 1 + 2t,\ 2 - t,\ 3 + t \rangle$ intersects $3x - y + 2z = 5$.

Angle Between Two Planes

Angle Between Planes

If planes have normals $\mathbf{n}_1$ and $\mathbf{n}_2$:

$$\cos\theta = \underline{\hspace{6cm}}$$

– Parallel: $\mathbf{n}_1 \parallel \mathbf{n}_2$ – Perpendicular: $\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$

Line of Intersection of Two Planes

Strategy

Given planes Π_1 and Π_2 with normals $\mathbf{n}_1$ and $\mathbf{n}_2$:

1. **Direction:** $\mathbf{d} = \underline{\hspace{3cm}}$
2. **Point:** Set one variable (e.g. $z = 0$) and solve the 2×2 system

Example 5: Intersection of Two Planes

Find the line of intersection of $2x - y + z = 3$ and $x + y - 2z = -4$.

4. Distances

Theorem 1: Distance from a Point to a Plane

Distance from $P_1 = (x_1, y_1, z_1)$ to $Ax + By + Cz = D$:

$$d = \underline{\hspace{6cm}}$$

Example 6: Point-to-Plane Distance

Find the distance from $Q = (1, -1, -3)$ to the plane $x + 2y + 3z = 18$.

5. Vector Projection onto a Plane

Projection onto a Plane

To project $\mathbf{a}$ onto a plane with normal $\mathbf{n}$:

$$\mathbf{a}_{\text{plane}} = \underline{\hspace{6cm}}$$

– Subtract the normal component from the vector

Example 7: Projection onto a Plane

Project $\mathbf{a} = \langle 3, 4, 5\rangle$ onto the plane with normal $\mathbf{n} = \langle 1, 1, 1\rangle$.

Show all work for full credit.

Practice Problems

Problem 1. Write the parametric equation of the line through $P = (2, 1, -3)$ in direction $\mathbf{d} = \langle 1, -2, 4\rangle$.

Problem 2. Find the equation of the plane through $(3, 0, -1)$ with normal $\mathbf{n} = \langle 2, 1, -5\rangle$.

Problem 3. Find where the line $\mathbf{r}(t) = \langle -t,\ 2t,\ t+3\rangle$ intersects the plane $-x + y = 6$.

Problem 4. Find the distance from $(2, 3, -1)$ to the plane $x - 2y + 2z = 4$.

Problem 5. Given $\Pi_1 : x + 2y - z = 3$ and $\Pi_2 : 2x - y + 3z = 1$: (a) find the angle between the planes, (b) find the line of intersection.

Answer (a): $\theta =$ ______________________________

Answer (b): $\mathbf{r}(t) =$ ______________________________